

# Pay a foreign invoice in one transaction.

Cross-border settlement on Arc at the FX rate that actually exists — built on the price oracle  
Arc launches without.

## THE PROBLEM

# A supplier bills you €5,000. You hold dollars.

**40–60bp**

through a payments provider, one to three days

**200–300bp**

through a bank, with nothing linking it back to the invoice

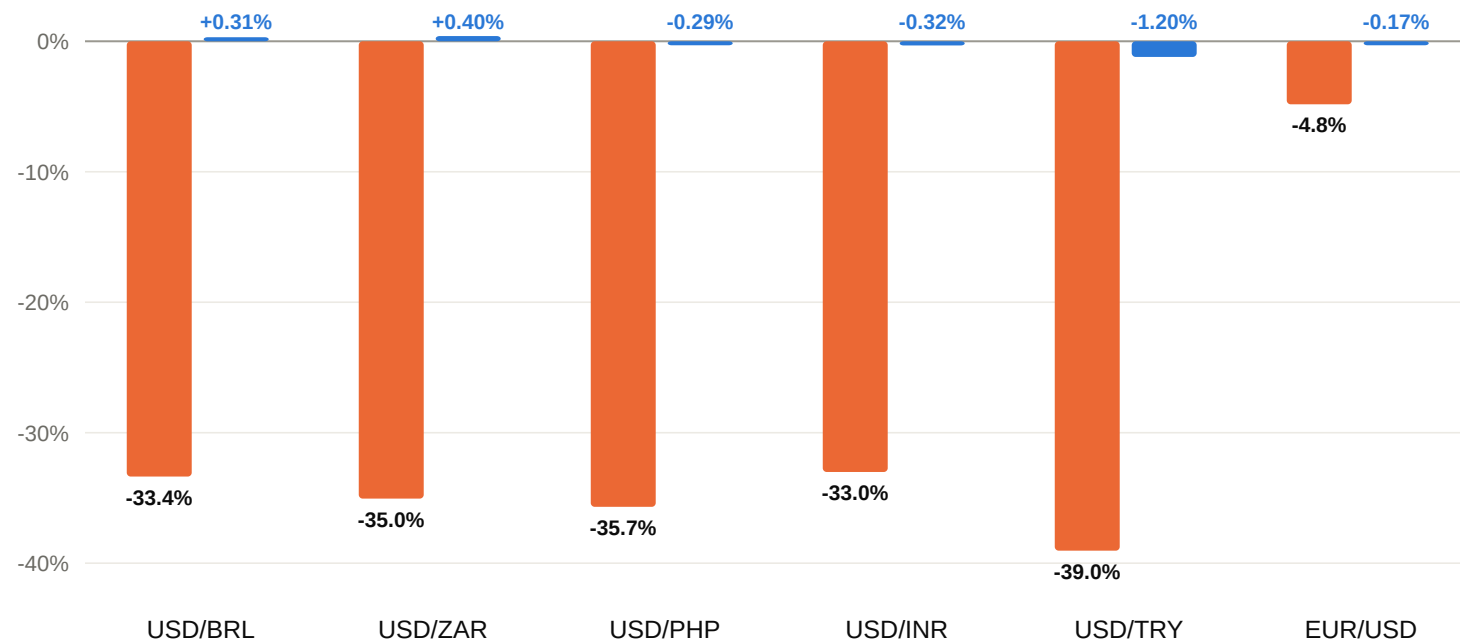
**0**

price oracles deployed on Arc, two days before mainnet

Circle has announced partner stablecoins in BRL, MXN, PHP, ZAR, JPY, KRW, CAD and AUD — none trading near 1.00. The only venue design on Arc is a stableswap that is cheapest at exactly 1.00, because a pool cannot centre on a rate the chain cannot read.

MEASURED · 259 REAL ECB OBSERVATIONS · SIX CORRIDORS

# What a pool with no oracle loses in a year



## LP return, per year

Two pools identical in every respect except where their liquidity sits. Same invariant, amplification, fee and trades.

**A third of the LP's capital, handed to arbitrageurs.**

Reproducible: npm run corridors

■ 1:1 stableswap ■ Rate-anchored

# Five contracts, deployed and source-verified on Arc testnet

- 1 RialtoOracle**

Pull-based FX feed. Publishers sign EIP-712 attestations; the caller names its own staleness bound; a circuit breaker caps any print at 10% fresh, 25% ever.
- 2 RialtoPool**

Stableswap on rate-scaled balances — the cheap part of the curve sits on the real rate and moves with it. 25bp, half to LPs, half to the protocol, in code.
- 3 RialtoSettlement**

Exact-output invoice settlement: denominated in what the payee receives, paid to a third party, reference and rate emitted on-chain.
- 4 RialtoRouter**

Multi-hop exact-output:  $N$  corridors give  $N(N-1)/2$  pairs.
- 5 RialtoForward**

Collateralised, cash-settled rate locks for invoices with terms.

DEMONSTRATED · EVERY HASH OPENS ON ARCSCAN

# Live on Arc testnet. Not a diagram.

**EUR 1.00**

received by the payee, exactly as billed

**3.00 USDC**

paid by a settled EUR 1,000 forward — exactly the rate move

**27bp**

all-in: 25bp fee + 2bp slippage on a EUR 20 book

**0.001253 EURC**

protocol revenue from that invoice, on-chain in protocolFees1

Oracle updating every 10 minutes unattended · six contracts verified · payer, payee and LP three separate addresses

# How it makes money

25bp

the fee — half of Wise, an order of magnitude under a bank

12.5bp

of everything settled goes to the protocol; the split is capped at half by the contract

12.53bp

what the testnet invoice actually paid — read off the pool, not projected

| DAILY SETTLED VOLUME            | ANNUAL PROTOCOL REVENUE |
|---------------------------------|-------------------------|
| \$250k                          | ~\$122,000              |
| \$1M (≈200 invoices of \$5,000) | ~\$489,000              |
| \$10M                           | ~\$4,890,000            |

The volume is new: an invoice settled here is a wire that would otherwise have gone through correspondent banking.

# Which part of the payment this is

## Getting in — not us

Circle Mint, an exchange, or a USDC balance the business already holds.

## The FX leg — this

USDC to the exact euro amount billed, at the real rate, 25bp, one transaction, invoice reference on-chain. Nothing on Arc could do this.

## Getting out — not us

The payee holds EURC, or cashes to a local account through a partner — several are Circle grantees.

No KYC, no fiat ramps, no local payout, no licence to hold client money. Circle already funds that network; what it cannot do today is convert at a rate the chain can verify.

WHY NOW

# Arc mainnet: 16 September 2026.

On that date there is still no price oracle on Arc. The first partner-stablecoin pool that tries to quote will centre on 1.00 — because that is the only number available to it. A day-one problem, not a someday problem.

# Distribution without a network — concretely

## 1 The two FX teams on Arc need the oracle

ViFi and Lunex run pools that cannot read a price. Offer the feed publicly; ask Circle for the introduction. Weeks 1–4.

## 3 Launch week

Arc ecosystem directory; the "Arc has no oracle" measurement as a technical post with the harness attached. Weeks 0–2.

## 2 Circle's payment grantees are the integrators

Hurupay, Blockradar, Payrit move USDC for customers paid in a second currency. A one-page settlement offer plus the live sandbox. Weeks 2–8.

## 4 India, when the INR stablecoin ships

Largest recipient of cross-border service payments; USD/INR already simulated at –33%/yr for 1:1 pools.

The first customers of an FX primitive are protocols and payment apps — reached through an address and an ABI, which one person can produce and has.

# Milestones — \$100,000 USDC over 24 weeks

| # | DELIVERABLE                                                                                                      | METRIC                                                               | WK | USDC   |
|---|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|----|--------|
| 1 | Oracle + USDC/EURC on Arc mainnet in launch week; 2-of-3 publishers on an intraday source; settlement front end  | Verified mainnet addresses; 14 days publishing; 100 settled invoices | 4  | 20,000 |
| 2 | A corridor per shipped partner stablecoin — BRL, MXN, PHP, ZAR first; CCTP v2 / Gateway; settlement API + SDK    | ≥4 corridors; \$500k settled; ≥3 external publishers                 | 10 | 25,000 |
| 3 | Oracle as public infrastructure: independent publishers, any pair on request, integration guide, security review | ≥2 external protocols reading it; ≥8 pairs                           | 16 | 25,000 |
| 4 | Depth: LP programme, payment-app routing, Circle Wallets, published corridor economics                           | \$5M settled; ≥25 paying businesses; ≥2 apps                         | 24 | 30,000 |

Milestone 1 alone is a coherent deliverable. Milestone 3 is the one worth arguing for: every lending market, perp and prediction market on Arc needs a feed, and none exists.

# I build systems a stranger can re-derive. Then I attack them.

SHIPPED BEFORE RIALTO

|                        |                                                                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Germline               | Verifiable AI-configuration tuning, live on <b>0G mainnet</b> ; every step recorded on-chain.                                                            |
| BIP 360 conformance    | From-scratch implementation vs the official reference, 13 vectors and 2,000 random trees: <b>two real gaps found in the reference.</b>                   |
| rein                   | A smart account an agent can operate and cannot drain. Verified source; a fully compromised agent tried six ways to empty it and lost nothing. 46 tests. |
| proofflow              | Revenue-based credit for emerging-market SMBs on Solana, built on <b>USDC and CCTP</b> , Arcium MPC. Live.                                               |
| economic-immune-system | Transaction authorisation against a budget policy, built on <b>Circle Developer Controlled Wallets.</b>                                                  |

So CCTP and Circle Wallets — marked *planned* in the milestones — are products I have already shipped with. What is new is Arc, and the primitive Arc is missing.

RIALTO · NINE DAYS · ONE PERSON

|                                       |                                        |
|---------------------------------------|----------------------------------------|
| 6                                     | 71                                     |
| contracts, source-verified on Arc     | passing tests                          |
| 259                                   | 37.8 → 3.2%                            |
| days of ECB data, six corridors       | oracle failure mode, found and bounded |
| 1 + 1                                 | 4:34                                   |
| invoice and forward, settled on-chain | demo video, reproducible from the repo |

Before all of it: Drape, built for an XPRIZE submission, and its postmortem — "built for a judge, not a shop". Rialto was built the other way round.